2-Month Machine Learning Execution Tracker (CS229 + Industrial Projects)

This plan combines CS229 theoretical learning with industrial-grade ML projects. Each day includes: CS229 focus, project work, and deliverable.

Week 1: ML Foundations + Sensor Anomaly System

- Day 1: Linear regression intuition + setup repo + define sensor anomaly problem
- Day 2: Gradient descent implementation + synthetic sensor data generation
- Day 3: Loss functions + baseline threshold anomaly detector
- Day 4: Regularization + Isolation Forest model
- Day 5: Evaluation metrics + anomaly scoring pipeline
- Day 6: Compare rule-based vs ML anomaly detection
- Day 7: Mini review + GitHub cleanup + visualization dashboard

Week 2: Logistic Regression + Predictive Maintenance Start

- Day 8: Logistic regression theory + NASA dataset setup
- Day 9: Feature engineering time-series signals
- Day 10: Train logistic regression model
- Day 11: Regularization tuning + evaluation
- Day 12: Start RUL prediction pipeline
- Day 13: Baseline RUL model (simple regression)
- Day 14: Documentation + results analysis

Week 3: Bias-Variance + Model Improvement

- Day 15: Bias-variance theory + cross-validation
- Day 16: Model selection experiments
- Day 17: XGBoost for RUL prediction
- Day 18: Hyperparameter tuning
- Day 19: Error analysis + failure cases
- Day 20: Improve feature engineering
- Day 21: Project consolidation

Week 4: SVM + Object Tracking Start

- Day 22: SVM intuition + decision boundaries
- Day 23: Implement SVM classifier

- Day 24: Start tracking system design
- Day 25: Kalman filter basics
- Day 26: Hungarian matching implementation
- Day 27: Multi-object tracking pipeline
- Day 28: Visualization system

Week 5: Probabilistic ML

- Day 29: Naive Bayes + probabilistic thinking
- Day 30: Gaussian models
- Day 31: Uncertainty estimation in RUL model
- Day 32: Tracking system improvement
- Day 33: Noise modeling in sensors
- Day 34: Add confidence scoring
- Day 35: Weekly integration review

Week 6: EM + Drift Detection System

- Day 36: EM algorithm intuition
- Day 37: Clustering experiments
- Day 38: Start drift detection system
- Day 39: Feature distribution shift detection
- Day 40: Model drift metrics
- Day 41: Stream processing design
- Day 42: Dashboard prototype

Week 7: Neural Networks + System Integration

- Day 43: Neural network basics + backprop intuition
- Day 44: Build simple NN model
- Day 45: Integrate anomaly + tracking systems
- Day 46: Drift system integration
- Day 47: System testing
- Day 48: Performance evaluation
- Day 49: Debug + optimization

Week 8: Deployment + Final Portfolio

- Day 50: FastAPI deployment setup
- Day 51: Dockerization (optional)
- Day 52: Streamlit dashboard finalization

- Day 53: System integration testing
- Day 54: End-to-end demo preparation
- Day 55: GitHub polishing + README
- Day 56: Final review + portfolio packaging